

PROJECT REPORT

Course Title : UI/UX Design

Course Code : CSE441

HydraFlow

Hydration & Electrolyte Dynamic Forecaster

A smart water-bottle companion interface that adapts hydration recommendations to weather, activity, sweat rate, and schedule.

Submitted By

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Abstract

HydraFlow is a mobile-first UI/UX concept for a smart water-bottle companion that provides context-aware hydration guidance. Instead of relying on fixed reminders, the proposed experience considers environmental conditions such as temperature and humidity, user activity and sweat rate, scheduled high-demand activities, and smart-bottle measurements. The interface translates these inputs into a daily hydration target, a recommended next drink, an explainable forecast, device status, and weekly insights.

The project focuses on making dynamic hydration information understandable and actionable. The prototype uses a guided onboarding flow followed by five primary areas: Home, Forecast, Smart Bottle, Insights, and Profile. This report documents the design rationale, user flow, information architecture, visual design decisions, prototype behavior, evaluation plan, limitations, and future improvements.

Keywords

UI/UX Design, Hydration, Smart Bottle, Context-Aware Interface, Personalized Recommendation, Mobile Application, Weather-Aware Design, Information Architecture

1. Introduction

1.1 Project Overview

HydraFlow is a UI/UX project that explores how a digital companion can make hydration management more adaptive and understandable. The central idea is a smart water-bottle ecosystem where the application does not merely remind a person to drink water; instead, it estimates changing hydration demand from multiple contextual signals and communicates the recommendation in a simple way.

1.2 Background

Hydration demand can vary with environmental heat, humidity, physical activity, individual characteristics, and the timing of demanding activities. A static reminder can therefore be poorly aligned with the user's actual routine. HydraFlow addresses this interaction problem by designing a single interface where these signals are converted into a clear daily target and timely actions.

1.3 Problem Statement

Users may know that hydration is important but still struggle to decide how much to drink, when to drink, and why their target should change. Existing reminder-style experiences can become repetitive because they often treat hydration as a fixed daily quantity. HydraFlow proposes a more contextual interaction model in which the interface explains changes and prioritizes the next useful action.

1.4 Project Scope

- Design a high-fidelity mobile interface for adaptive hydration guidance.
- Create an onboarding flow that collects the contextual information needed for personalization.
- Represent weather, activity, schedule, bottle status, and hydration history in understandable visual forms.
- Design an interactive prototype suitable for live demonstration and usability testing.
- Define an evaluation framework for task completion, clarity, efficiency, and navigation.

1.5 Project Deliverables

- User flow and information architecture
- High-fidelity mobile UI screens
- Interactive Figma prototype / website
- Design system and reusable interface components
- Usability evaluation plan
- Final project presentation and report

2. Objectives and Requirements

2.1 General Objective

To design a user-centered smart hydration companion that communicates personalized hydration recommendations clearly, efficiently, and with sufficient explanation for users to understand why their target changes.

2.2 Specific Objectives

- Reduce the effort required to understand the user's current hydration status.
- Provide a personalized daily hydration target rather than a purely static reminder.
- Use weather, activity, sweat-rate, and schedule information as contextual factors in the interface.
- Provide a clear “next drink” recommendation so the user knows what to do next.
- Make the recommendation explainable through a “Why your target changed” interaction.
- Show smart-bottle information such as current level, temperature, and battery status.
- Provide weekly insights so users can review consistency and progress.
- Maintain a coherent navigation structure across Home, Forecast, Bottle, Insights, and Profile.

2.3 Functional Requirements

ID	Area	Requirement
FR-01	Onboarding	The system shall guide first-time users through setup steps.
FR-02	Bottle	The interface shall allow users to connect or skip smart-bottle setup.
FR-03	Lifestyle	The interface shall capture an activity level for baseline estimation.
FR-04	Weather	The interface shall request location permission for weather-aware recommendations.
FR-05	Schedule	The interface shall represent scheduled high-demand activities.
FR-06	Home	The home screen shall display the current target, progress, demand, and next drink.
FR-07	Forecast	The forecast screen shall visualize predicted demand across time.
FR-08	Bottle status	The bottle screen shall show current level, temperature, battery, and sync status.

FR-09	Insights	The insights screen shall summarize weekly intake and goal consistency.
FR-10	Profile	The profile shall expose personal and hydration-related settings.

3. Target Users and User Needs

3.1 Primary User Persona

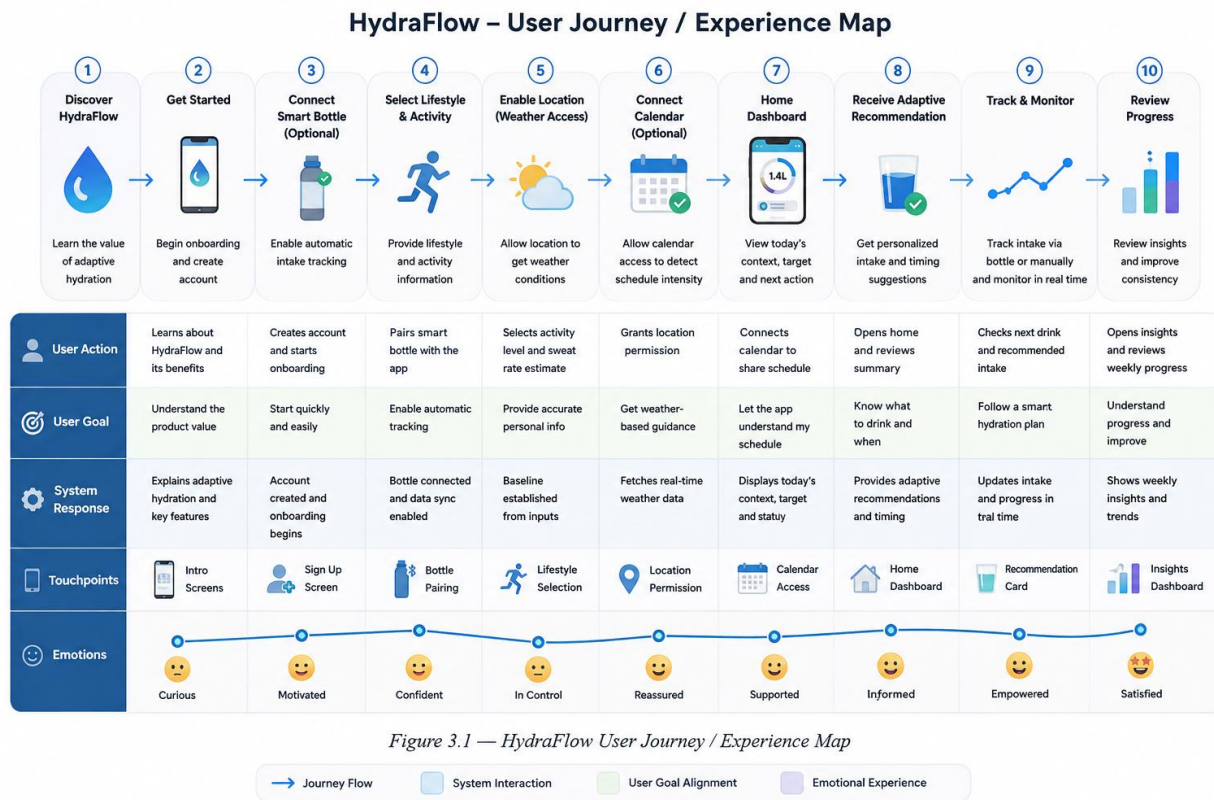
Persona	[Persona Name]
Profile	Student / young professional with a busy daily schedule
Routine	Classes or work, commuting, and occasional exercise
Pain Points	Forgets to drink; does not know how much is enough; routine changes throughout the day
Needs	Fast guidance, clear progress, contextual recommendations, low-effort logging

3.2 User Needs

- Understand the current hydration state at a glance.
- Know the recommended amount and timing of the next drink.
- Understand why the target has changed.
- See how upcoming activities may affect hydration demand.
- Confirm that the smart bottle is connected and transmitting data.
- Review progress without needing to interpret complicated analytics.

3.3 User Journey Summary

The journey begins with a short onboarding sequence. The user first learns the product value, optionally connects the smart bottle, provides lifestyle information, grants location access, and connects a schedule. Once setup is complete, the user lands on the Home dashboard where the system summarizes today's context and provides an actionable recommendation.



4. Design Methodology

4.1 User-Centered Design Approach

The project follows a user-centered design mindset in which the interface is organized around user questions rather than around raw data sources. The primary question is: “What should I do now?” Supporting questions are: “How much do I need today?”, “Why did it change?”, “What is coming next?”, and “How am I doing over time?”

4.2 Design Process

1. Problem definition
2. User and requirement analysis
3. Information architecture
4. User flow design
5. Low-fidelity exploration
6. High-fidelity interface design
7. Interactive prototyping
8. Usability review and refinement

4.3 Design Principles

Principle	Application in HydraFlow
Action first	The Home screen prioritizes the next drink and current progress.
Progressive disclosure	Detailed explanations are available without overwhelming the first view.
Explainability	Target changes are accompanied by understandable factors.
Consistency	Navigation, buttons, cards, typography, and status patterns are reused.
Accessibility	Contrast, touch targets, labels, and readable hierarchy are considered.
Context awareness	Weather, activity, schedule, and bottle data are presented as connected signals.

5. Information Architecture and User Flow

5.1 Application Structure

The application is organized around five primary destinations accessible through persistent bottom navigation.

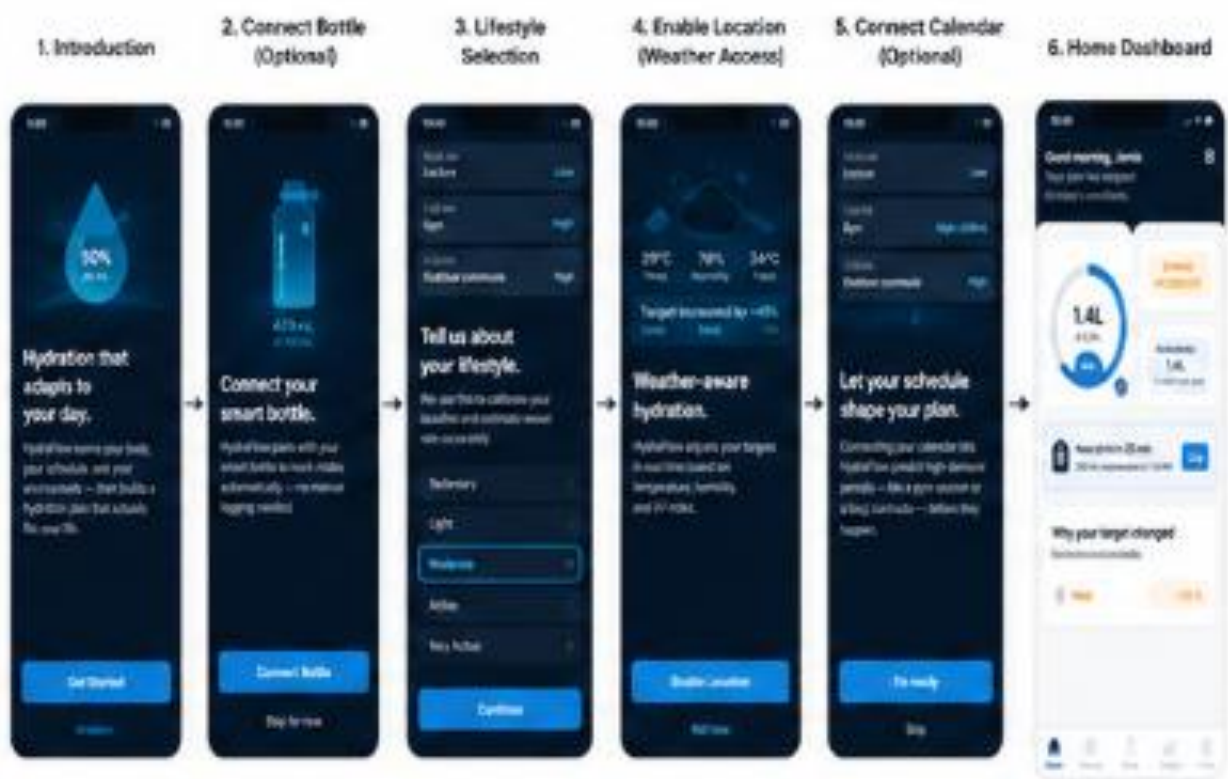
- Home — immediate hydration status and next action.
- Forecast — time-based predicted demand and environmental influence.
- Bottle — connected-device status and synchronization.
- Insights — historical performance, target achievement, and consistency.
- Profile — personal information, activity level, sweat rate, and preferences.



5.2 Onboarding Flow

9. Introduction → Get Started
10. Connect Smart Bottle → Connect Bottle / Skip
11. Lifestyle → Select Activity Level
12. Weather → Enable Location / Not Now
13. Schedule → Connect Calendar / Skip
14. Home → Personalized hydration dashboard

3. Figure 5.2 — Onboarding Flow



5.3 Core Task Flow

A representative task begins when the user opens Home. The user sees the current target and hydration progress, checks the next recommended drink, optionally opens the explanation for the target change, and can navigate to Forecast for deeper context or Bottle for device status.

6. Interface Design

6.1 Visual Direction

HydraFlow uses a health-and-technology visual language. A deep navy interface establishes a calm, focused environment, while cyan/blue accents communicate water, activity, and interactive states. White and pale surfaces are used in data-heavy areas to improve readability and separate analytical content from the darker navigation/header areas.

6.2 Typography

The prototype uses a clean sans-serif interface style with strong hierarchy between page titles, section headings, metric values, supporting descriptions, and labels. For this report, Times New Roman is used to maintain formal academic formatting.

6.3 Color and Status Language

Role	Visual Meaning	Example Use
Primary blue/cyan	Hydration, interaction, progress	Buttons, progress rings, selected navigation
Green	Positive/connected state	Bottle connected, battery/goal status
Orange	Attention / elevated demand	Heat, moderate/high demand
Neutral navy/gray	Background and secondary information	Headers, cards, supporting labels

6.4 Component Strategy

- Primary CTA buttons
- Navigation bar
- Metric cards
- Weather card
- Hydration progress ring
- Timeline/chart cards
- Status badges
- Selection controls
- Bottle status card
- Insight cards

7. Prototype Screens and Interaction Design

7.1 Onboarding Screens

Hydration Introduction

Communicates the core value proposition: hydration adapts to the user's day.

Smart Bottle Connection

Introduces automatic intake tracking and provides a skip path.

Lifestyle Selection

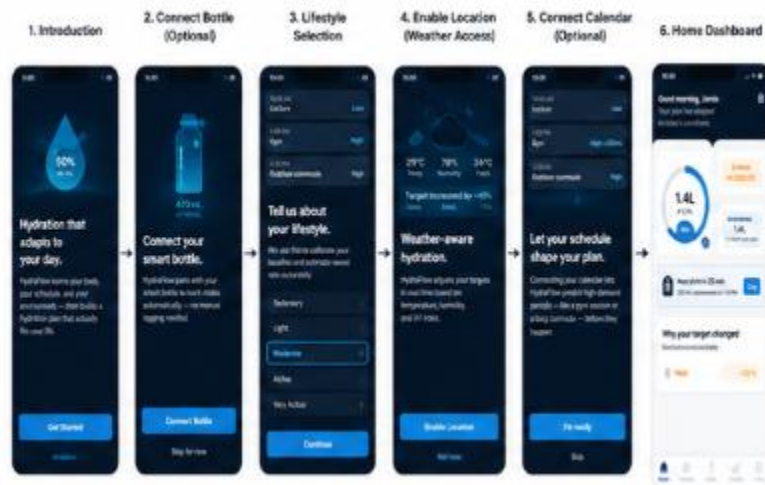
Captures activity level to calibrate baseline hydration needs.

Weather-Aware Hydration

Explains that temperature, humidity, and UV can influence the target.

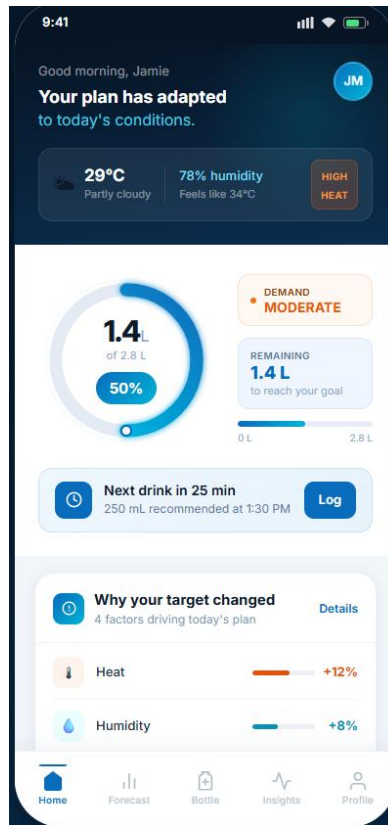
Schedule Connection

Shows how demanding events such as gym sessions or outdoor commutes can affect the plan.



7.2 Home Dashboard

The Home screen is the primary decision surface. It combines weather conditions, the current hydration target, progress, demand level, remaining amount, and the next recommended drink. A “Why your target changed” area provides an explanation rather than leaving the user with an unexplained number.



7.3 Forecast

The Forecast screen transforms the daily target into a time-based view. The timeline compares recommended, actual, and demand signals so users can understand when hydration demand is expected to peak.

8. Smart Bottle, Insights, and Profile

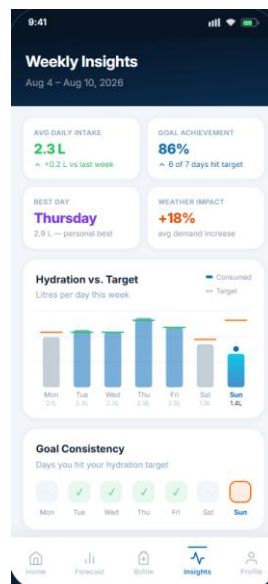
8.1 Smart Bottle

The Smart Bottle screen provides a direct view of the connected device. It communicates the current water level, capacity, temperature, battery status, and synchronization state. The goal is to reassure the user that physical bottle data is being captured and to reduce uncertainty around device connectivity.



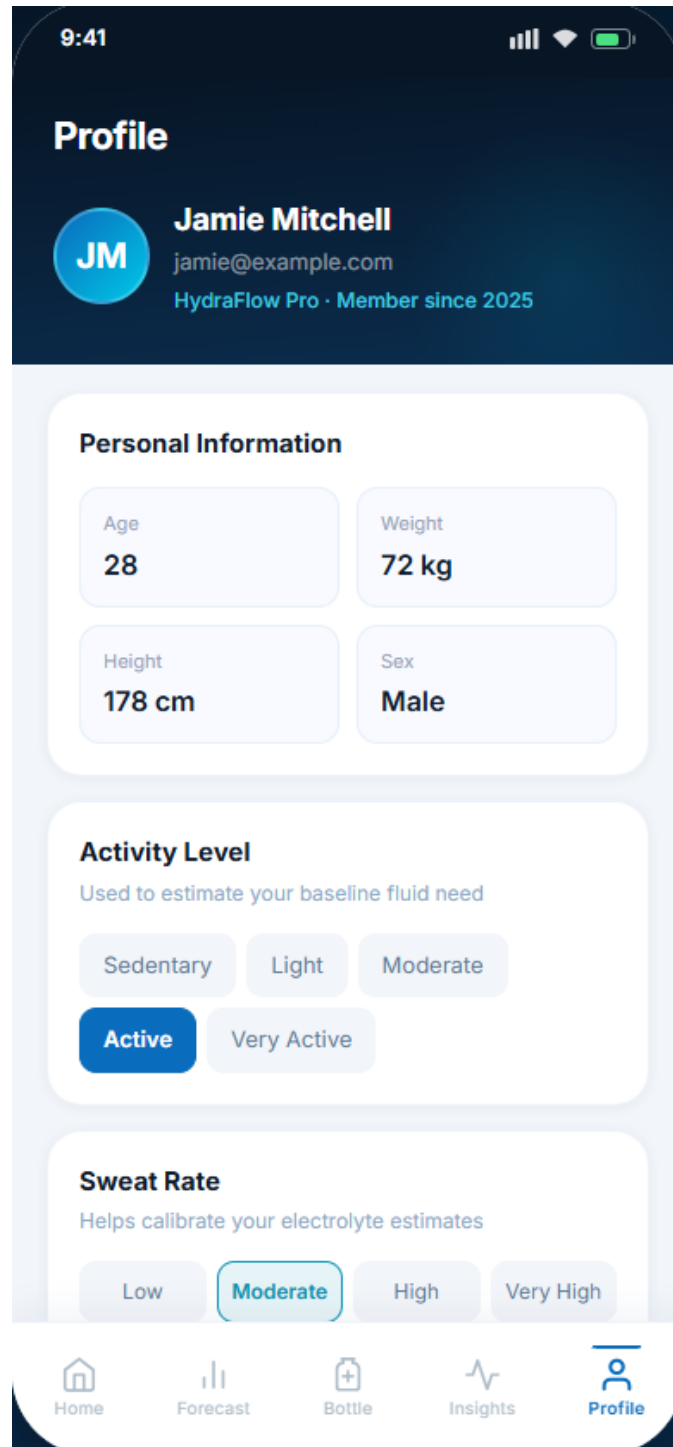
8.2 Weekly Insights

The Insights screen moves the experience from immediate action to reflection. It presents average daily intake, goal achievement, best day, weather impact, hydration-versus-target visualization, and goal consistency. The use of compact cards and charts helps the user scan progress without requiring complex analysis.



8.3 Profile

The Profile screen provides the controls needed to maintain personalization. It contains personal information, activity level, and sweat-rate selection. These controls are important because the recommendation logic depends on user-specific characteristics.



9. Usability Evaluation Plan

9.1 Evaluation Goals

The prototype should be evaluated to determine whether users can quickly understand the hydration recommendation, identify the next action, interpret the reason for changes, and navigate between the major product areas.

9.2 Proposed Test Tasks

15. Task 1: Find today's hydration target.
16. Task 2: Determine when the next drink is recommended.
17. Task 3: Explain why today's target changed.
18. Task 4: Find the period of highest expected hydration demand.
19. Task 5: Check the smart bottle's current water level and battery.
20. Task 6: Review weekly goal achievement.
21. Task 7: Change the activity or sweat-rate setting in Profile.

9.3 Evaluation Metrics

Metric	What It Measures	Suggested Target	Actual Result
Task completion rate	Whether users finish each task	$\geq 90\%$	[Enter result]
Time on task	Efficiency of key interactions	[Set target]	[Enter result]
Error rate	Navigation/interaction mistakes	Low	[Enter result]
Comprehension	Whether users understand recommendations	$\geq 80\%$	[Enter result]
Satisfaction	Perceived ease and usefulness	[Use SUS/UMUX-Lite]	[Enter result]

10. Accessibility and Inclusive Design

- Maintain sufficient contrast between text and background.
- Use text labels alongside color-based status indicators so meaning is not dependent on color alone.
- Provide clear labels for buttons and controls.
- Keep primary touch targets large enough for comfortable mobile interaction.
- Use concise language and a clear information hierarchy.
- Do not rely only on charts; provide supporting labels and values.
- Provide a visible alternative to optional permissions such as location, bottle connection, and schedule connection.

10.1 Accessibility Checklist

Area	Check	Status
Color contrast	Text and controls remain readable	[To verify]
Typography	Body text remains legible on mobile	[To verify]
Touch targets	Primary controls are easy to tap	[To verify]
Labels	Icons have meaningful labels	[To verify]
Charts	Important information is not chart-only	[To verify]
Permissions	Skip/not-now alternatives exist	[Implemented / Verify]

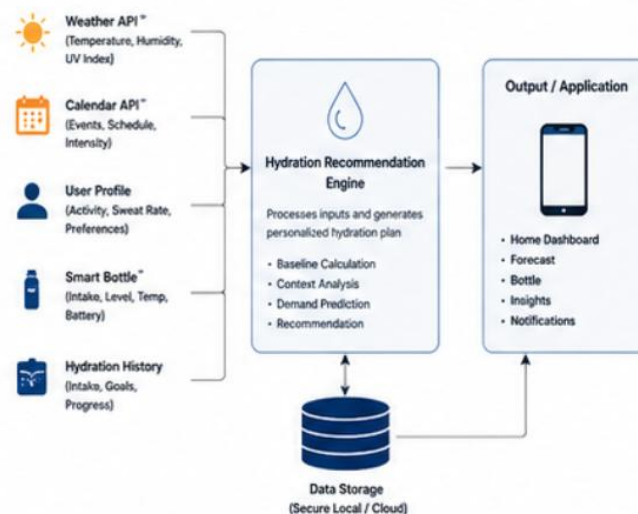
11. Technical and System Feasibility

Although HydraFlow is presented primarily as a UI/UX prototype, its information architecture is designed around data sources that could be integrated into a real system. The interface separates data acquisition, recommendation logic, and presentation so that future implementation can evolve without redesigning the entire user experience.

11.1 Proposed Data Inputs

- Weather API: temperature, humidity, UV and related conditions.
- Calendar integration: event timing and activity intensity.
- User profile: activity level, body-related inputs, and sweat-rate estimate.
- Smart bottle: water level, consumption, temperature, battery, and synchronization.
- Hydration history: intake and goal-achievement records.

11.2 Conceptual System Flow



11.3 Engineering Considerations

- Use secure permission handling for location and calendar access.
- Treat hydration recommendations as guidance rather than medical diagnosis or treatment.
- Allow users to edit or override inaccurate profile information.
- Handle missing or stale sensor/API data gracefully.
- Provide clear connection and synchronization states.
- Design for intermittent connectivity and delayed device synchronization.

12. Limitations and Future Work

12.1 Current Limitations

- The current deliverable is a UI/UX prototype rather than a production medical or consumer hydration system.
- Real-time weather, calendar, and bottle integrations may be represented conceptually in the prototype.
- Recommendation values shown in the prototype should not be interpreted as clinically validated hydration prescriptions.
- Usability metrics are placeholders until formal user testing is conducted.
- The current prototype may not cover every edge case such as sensor failure, missing permissions, or conflicting schedule events.

12.2 Future Improvements

- Integrate live weather and calendar services.
- Connect to a real smart-bottle Bluetooth/IoT data source.
- Develop a validated recommendation model using appropriate domain expertise and evidence.
- Add intelligent notification timing that avoids notification fatigue.
- Introduce personalized trend learning from historical intake and activity.
- Expand accessibility testing with users with different abilities.
- Conduct iterative usability testing and A/B comparison of key dashboard layouts.

13. Conclusion

HydraFlow demonstrates how a UI/UX design can transform a complex, context-dependent problem into a focused mobile experience. The project moves beyond static reminders by connecting weather, activity, sweat rate, schedule, and smart-bottle information to an adaptive hydration plan.

The strongest design decision is the separation between recommendation and explanation. The Home screen tells the user what to do, while Forecast, Insights, Bottle, and Profile provide the supporting context needed to trust and manage the system. The resulting experience is structured around a simple interaction principle: understand the user's context, communicate the next useful action, and explain the reason behind it.

With further usability testing and real integrations, the HydraFlow concept can be developed into a more complete smart hydration ecosystem while maintaining the same user-centered foundation.

Project Prototype

Figma prototype / live website: <https://distort-salt-54882236.figma.site/>

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27. [6] [Smart-bottle / IoT source used by the project].
28. [7] [Any academic paper, article, or design reference actually used by the team].

Appendix A — Prototype Screen Inventory

ID	Screen	Section
A1	Hydration Introduction	Onboarding
A2	Connect Smart Bottle	Onboarding
A3	Lifestyle Selection	Onboarding
A4	Weather-Aware Hydration	Onboarding
A5	Schedule Connection	Onboarding
A6	Home Dashboard	Main App
A7	Hydration Forecast	Main App
A8	Smart Bottle	Main App
A9	Weekly Insights	Main App
A10	Profile	Main App

Appendix B — Items to Replace Before Submission

- Project title if required by the course
- Course title and course code
- Student names and IDs
- Instructor name and designation
- Department / section / semester
- Submission date
- Final user persona and research findings
- Final screenshots and diagrams in all figure placeholders
- Actual usability testing results
- Final references and URLs
- Final Figma prototype URL